

ont_asm_caller: statistical models for allele-specific methylation from single-individual long-read data

Method and theory notes

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Abstract

Long-read sequencing assigns individual molecules to a haplotype, making allele-specific methylation (ASM) callable in a *single individual* with no biological replicates. This document sets out the models in `ont_asm_caller`, the theory behind them, the generative simulators used to validate them, and what the first real read-level measurement on this project's data says about which of them is actually needed. Three testing paths are described: a coverage-aware beta-binomial likelihood-ratio test on counts pooled across spatially clustered CpGs; a read-level test that treats the molecule as the unit of observation; and a read-*pattern* test that models the joint methylation configuration along a molecule. The quantity that decides among them is the *design effect*, the factor by which sharing molecules across CpGs inflates the apparent sample size. We derive it, show it is attenuated by basecalling error in a way that admits an exact correction, and report its first measurement here: on HG00146 chr15 the median design effect at the caller's own region scale is 1.29, with only 0.7% of regions above 5. That is a substantially more benign number than the simulation-based critique that motivated the read-level and pattern paths had assumed, and it demotes them from replacements to fallbacks.

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1 The problem, and why the obvious test fails

1.1 Setting

For a single individual, reads are phased onto two haplotypes using heterozygous SNPs. At CpG locus l and haplotype $h \in \{1, 2\}$ let $N_{l,h}$ be the number of reads covering l and assigned to h , and $X_{l,h}$ the number of those reads carrying a methylated call. The question at each locus is whether the two haplotypes' underlying methylation rates differ.

The defining constraint is that **there are no replicates**. A haplotype is one half of one individual's reads, not an independent biological sample. Tools built for differential methylation between *groups* (DSS, methylKit) estimate their overdispersion from between-replicate variance, a quantity that is not merely noisy here but undefined. On this project's data DSS's dispersion step ran indefinitely without output, which is consistent with hitting exactly that degenerate case.

1.2 Fisher's exact test is depth-confounded

The natural per-locus test is a 2×2 Fisher exact test on (methylated, unmethylated) $\times$ ($h=1, h=2$). Run on an 18-donor HPRC cohort it produced, on the same chromosome at identical thresholds, **113,119** significant loci in one sample and **2** in another: a $> 5 \times 10^4$ ratio. The two samples' *raw* effect sizes were similar. What differed was per-haplotype depth (median $18\times$ vs. $8\times$).

The mechanism is that Fisher's test conditions on the margins and models only binomial sampling. It has no parameter for extra-binomial variation, so as depth grows its p -values become more confident rather than better calibrated. A procedure whose output tracks coverage cannot be

compared across samples, and real long-read cohorts always vary in coverage. In simulation against a known ground truth, 77–90% of Fisher’s calls were false positives (§9). This is the defect the package exists to fix, and it is not in question; everything contested below is about what should *replace* it.

2 Path A: the coverage-aware beta-binomial model

2.1 Model

At locus l , haplotype h :

$$X_{l,h} \mid N_{l,h}, p_{l,h} \sim \text{Binomial}(N_{l,h}, p_{l,h}), \quad (1)$$

$$p_{l,h} \sim \text{Beta}(\mu_{l,h}, \rho), \quad (2)$$

parameterised by mean μ and dispersion ρ , i.e. shape parameters $a = \mu(1 - \rho)/\rho$ and $b = (1 - \mu)(1 - \rho)/\rho$. Marginally $X \sim \text{BetaBinomial}(N, \mu, \rho)$ with

$$\text{Var}(X) = N\mu(1 - \mu)[1 + (N - 1)\rho]. \quad (3)$$

Equation (3) is the whole point: variance grows super-linearly in depth, so a deeply covered locus is not automatically a confidently called one. Depth enters the likelihood directly, which is what makes a single threshold meaningful across samples of different coverage.

2.2 Test

With ρ fixed at a pre-estimated value, each locus gets a likelihood-ratio test of $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$,

$$\Lambda = 2 \left[\max_{\mu_1} \ell(\mu_1) + \max_{\mu_2} \ell(\mu_2) - \max_{\mu} \{\ell_1(\mu) + \ell_2(\mu)\} \right] \stackrel{H_0}{\sim} \chi_1^2. \quad (4)$$

Each maximisation is a bounded one-dimensional problem (`model.fit_mu`).

2.3 Where the dispersion comes from: loci instead of replicates

This is the substantive modelling choice. There are no replicates, but there are *millions of loci*. Under the assumption that the large majority of loci carry no true ASM, $X_{l,1}$ and $X_{l,2}$ are, under the null, two draws from a shared rate — exchangeable in exactly the way replicates would be, found across the genome rather than across samples. Pooling that excess variance gives a well-defined global ρ .

The device is not new; it is standard in allele-specific *expression*, where the same “one individual, two alleles, no replicates” structure arises: MBASED (Mayba et al. 2014), QuASAR (Harvey et al. 2015) and RASQUAL (Kumasaka et al. 2016) all estimate dispersion by pooling across loci. `ont_asm_caller` ports that machinery to methylation counts.

Concretely (`dispersion.estimate_dispersion`), with $\hat{p}_l = (X_{l,1} + X_{l,2})/(N_{l,1} + N_{l,2})$,

$$\chi_l^2 = \sum_{h=1,2} \frac{(X_{l,h} - N_{l,h}\hat{p}_l)^2}{N_{l,h}\hat{p}_l(1 - \hat{p}_l)}, \quad \mathbb{E}[\chi_l^2] \approx 1 + \rho(\bar{n}_l - 1), \quad (5)$$

with $\bar{n}_l = (N_{l,1} + N_{l,2})/2$, and ρ fitted by least squares of $\chi_l^2 - 1$ on $\bar{n}_l - 1$ through the origin.

2.4 A boundary failure, and the mean-dependent fix

The estimator above assumes $\mathbb{E}[\chi_l^2] = 1$ under pure binomial sampling. That is asymptotic and it fails at the boundaries. When $X_{l,1} = N_{l,1}$ and $X_{l,2} = N_{l,2}$ — both haplotypes fully methylated, the single most common configuration in a real bimodal methylome — we get $\hat{p}_l \rightarrow 1$, the variance term collapses, $\chi_l^2 \rightarrow 0$, and the locus contributes evidence for $\rho = 0$. Measured bias in recovering a known $\rho = 0.05$:

μ distribution	true ρ	$\hat{\rho}$	ratio
all $\mu = 0.5$	0.05	0.0493	0.99
uniform(0.2, 0.8)	0.05	0.0489	0.98
realistic bimodal	0.05	0.0273	0.55
all $\mu = 0.95$	0.05	0.0180	0.36

Underestimating ρ makes the test anti-conservative. Worse, the original benchmark used $\mu = 0.5$ everywhere, which is precisely the one regime where the estimator is unbiased — so the validation could not have detected this.

Beta-binomial dispersion is also structurally mean-dependent: bounded near $\mu = 0$ and 1, largest near $\mu = 1/2$. A single global ρ is simultaneously too conservative at the boundaries (where imprinting-style ASM lives) and too liberal in the middle. `estimate_dispersion_trend` replaces the $\mathbb{E}[\chi^2] = 1$ assumption with a simulated baseline at the observed $(N_1, N_2, \hat{p})$ — reproducing the same boundary degeneracy — and reads ρ off the excess, fitted in bins of $\hat{p}$ and returned as a callable $\rho(\mu)$. It overshoots by 2–3 $\times$ at $\mu = 0.05, 0.95$. Overshoot is conservative, so the failure mode is safe, but it costs power exactly where imprinting-style ASM lives.

3 Region pooling, and the design effect

3.1 Why pool

The per-CpG test of §2 is correctly calibrated in isolation and close to useless genome-wide. At a realistic true-ASM prevalence ($\sim 0.5\%$, the deCODE Icelandic validated rate) across millions of tests, a true positive’s raw p -value must be extraordinarily small to survive Benjamini–Hochberg, and per-CpG depth cannot produce values that small. Measured power after BH: 0.3% at 18 $\times$.

Pooling counts across CpGs within 500 bp (`region.cluster_cpgs`, matching this project’s `GAP_MERGE_BP` and DSS’s `smoothing.span`) attacks both halves of the problem: fewer tests, so a looser BH threshold; and genuinely lower per-test noise, since averaging several site-level draws cancels some site-specific dispersion. A `max_span` cap is needed because a bare adjacent-gap rule daisy-chains through CpG-dense stretches — on real chr1 data, without a cap, the largest “region” reached 21,163 CpGs spanning 335 kb.

A non-obvious consequence: pooling several independent per-CpG beta-binomial draws *lowers* the effective dispersion of the combined count, so reusing the per-CpG $\hat{\rho}$ for pooled counts is misspecified and needlessly conservative. Dispersion must be re-estimated on the pooled counts themselves.

3.2 The design effect: definition and decomposition

Pooling assumes the pooled observations are independent. For long reads they are not, and the amount by which they are not is the *design effect*. This is the quantity the user asked about, so we derive it.

Consider a region with C CpGs and R reads on one haplotype, where every read covers every CpG — which is what a 35 kb molecule does to a 500 bp window. Let $X_{rj} \in \{0, 1\}$ be read r ’s call at CpG j , with $\mathbb{E}[X_{rj}] = \mu$, $v = \mu(1 - \mu)$, and within-read correlation $\text{Corr}(X_{rj}, X_{rk}) = r_{jk}$ for $j \neq k$. Define the per-read methylation fraction $F_r = C^{-1} \sum_j X_{rj}$. Then

$$\text{Var}(F_r) = \frac{1}{C^2} \left[Cv + \sum_{j \neq k} v r_{jk} \right] = \frac{v}{C} \left[1 + (C - 1) \bar{r} \right], \quad \bar{r} = \binom{C}{2}^{-1} \sum_{j < k} r_{jk}. \quad (6)$$

Under independence the fraction would have variance v/C , so

$$\boxed{\text{DE} = \frac{\text{Var}(F_r)}{v/C} = 1 + (C - 1) \bar{r}} \quad (7)$$

Naive pooling treats RC read×CpG observations as independent when the effective number is RC/DE . The likelihood then behaves as if it had DE times more information than it does, the LRT statistic inflates, and type I error breaks.

Equation (7) answers “what are the design-effect elements”: exactly two, the number of CpGs pooled C and their mean pairwise within-read correlation $\bar{r}$. Both are measurable. `readlevel.design_effect` estimates the left-hand side directly from per-read fractions; $\bar{r}$ is an integral of the co-methylation correlation function (§4) over the region’s CpG spacings, which is what `decay.implied_design_effect` computes. **Having two independent estimators of the same quantity is deliberate: their disagreement is the tripwire that says the measurement is not yet trustworthy.**

3.3 What it costs

Measured on null regions ($\mu_1 = \mu_2$ everywhere) with shared molecules, and with ρ re-estimated on the pooled counts exactly as prescribed:

within-read co-methylation	type I at $\alpha = 0.05$	type I at $\alpha = 0.001$
0.0 (the independence assumption)	0.052	0.0015
0.5	0.060	0.0013
1.0	0.144 (2.9×)	0.0112 (11×)

The inflation is worst in the extreme tail, which is exactly where genome-wide BH operates. It also worsens with region size, because a single global ρ cannot absorb a design effect that scales with C .

4 The co-methylation correlation function

$\bar{r}$ in (7) is an average of a distance-dependent correlation function $r(d)$ over the CpG–CpG distances present in a region. Measuring $r(d)$ is therefore the root measurement of this whole architecture — and it is the same object as the parent project’s methylation-LD decay curve, so one measurement settles both analyses.

4.1 Estimation

`decay.comethylation_decay` pools pairwise covariances into distance bins. Each CpG column is centred by *its own* within-region mean before pooling; pooling raw products across regions with different methylation levels would manufacture correlation out of between-region mean variation. Within bin b ,

$$\hat{r}_b = \frac{\sum_{\text{regions}} \sum_{(j,k) \in b} S_{xy}^{(jk)}}{\sqrt{\sum \sum S_{xx}^{(jk)} \cdot \sum \sum S_{yy}^{(jk)}}}, \quad (8)$$

a variance-weighted pooled correlation.

4.2 Basecalling error attenuates it, exactly

Methylation calls are not observed, they are inferred, and a wrong binary call biases every correlation toward zero. Write $S = 2X - 1 \in \{-1, +1\}$ and let the observed call be $\tilde{S} = S \cdot R$ where $R = -1$ with probability ε , independently across sites and reads, so $\mathbb{E}[R] = 1 - 2\varepsilon$. Then

$$\text{Cov}(\tilde{S}_j, \tilde{S}_k) = \mathbb{E}[S_j S_k] (1 - 2\varepsilon)^2 - \mathbb{E}[S_j] \mathbb{E}[S_k] (1 - 2\varepsilon)^2 = (1 - 2\varepsilon)^2 \text{Cov}(S_j, S_k), \quad (9)$$

while $\text{Var}(\tilde{S}_j) = 1 - (1 - 2\varepsilon)^2 m_j^2$ with $m_j = \mathbb{E}[S_j]$. Hence

$$\text{Corr}(\tilde{S}_j, \tilde{S}_k) = (1 - 2\varepsilon)^2 \text{Corr}(S_j, S_k) \cdot \frac{\sqrt{(1 - m_j^2)(1 - m_k^2)}}{\sqrt{(1 - (1 - 2\varepsilon)^2 m_j^2)(1 - (1 - 2\varepsilon)^2 m_k^2)}}. \quad (10)$$

At $\mu = 1/2$ (where $m_j = 0$) the correction factor is exactly $(1 - 2\varepsilon)^2$. Away from $\mu = 1/2$ the second factor is < 1 , so the true attenuation is slightly *stronger* than $(1 - 2\varepsilon)^2$ and dividing by $(1 - 2\varepsilon)^2$ **under-corrects** — the de-attenuated correlation is a lower bound, which is the safe direction.

Simulation at $\mu = 1/2$ confirms the law to three decimals, and shows the practical stakes. On data with a true design effect of 6.27:

ε	0.00	0.02	0.05	0.10	0.20
measured DE	6.27	5.85	5.25	4.36	2.89
fitted decay length (bp)	613	612	614	616	623

Call error moves the design effect across the entire range that decides whether pooling is legal, while leaving the decay *length* untouched. Only the amplitude is attenuated. This is why the extraction step must emit a no-call rather than a coin flip for ambiguous positions: a no-call costs data, a wrong call costs calibration, and (10) can only correct an ε that is unbiased.

5 Path B: the molecule as the unit of observation

If the problem with pooling is that molecules are counted C times, the direct fix is to summarise each molecule to one number and compare the two haplotypes' read-level distributions. Nothing then needs to model co-methylation: aggregating within a read absorbs it, and the between-read variance driving the test is estimated from the reads themselves. **No dispersion parameter is required at all.**

`readlevel.test_region_reads` takes per-read fractions F_r and applies a Welch t -test, alongside a Mann–Whitney test, keeping the more conservative of the two. Welch rather than pooled-variance because the two haplotypes routinely differ in both depth and spread; Mann–Whitney because read fractions are bounded in $[0, 1]$ and pile up at the boundaries under strong co-methylation, where t asymptotics are poor.

Complete separation. If every read on $h=1$ lies above every read on $h=2$ — the strongest possible ASM signal, e.g. a fully imprinted DMR — both within-group variances are zero and the t statistic is infinite. The exact randomisation p -value is the right answer: of the $\binom{n_1+n_2}{n_1}$ equally likely relabellings, only the observed split and its mirror are this extreme, giving $p = 2/\binom{n_1+n_2}{n_1}$. An earlier version floored this at $1/(n_1 + n_2) \approx 0.03$, which never survives BH, and that single bug cost essentially all power at high co-methylation.

Cost. The read-level test is conservative at every co-methylation level and never breaks. Its low power at high design effect is *correct*: at $DE = 5.5$ there genuinely are $\sim 5\times$ fewer independent observations, and the pooled test’s apparent power in that regime is not power, it is a 70% false discovery rate.

6 Path C: the read pattern

Reducing a molecule to a fraction discards the within-read *pattern*, and that pattern is the epiallele signal. The published precedent for using it is CPEL (Abante, Fang, Feinberg & Goutsias, *Nat Commun* 2020), which fits a 1-D Ising model per haplotype over $N \leq 20$ CpGs,

$$p(x) \propto \exp\left(\sum_k \alpha_k \sum_{n \in R_k} s_n + \beta \sum_n s_n s_{n+1}\right), \quad s_n = 2x_n - 1, \quad (11)$$

with one field α_k per ≤ 500 bp subregion and one nearest-neighbour coupling β , and compares the two fitted distributions via

$$T_{\text{MML}} = |\text{MML}_1 - \text{MML}_2| \quad \text{mean methylation imbalance,} \quad (12)$$

$$T_{\text{NME}} = |\text{NME}_1 - \text{NME}_2| \quad \text{entropy imbalance,} \quad (13)$$

$$T_{\text{PDM}} = \text{JSD}(p_1, p_2)/N \quad \text{whole-distribution imbalance,} \quad (14)$$

where $\text{NME} = H(X)/N$ is the normalised Shannon entropy. T_{NME} has no counterpart in Paths A or B: it detects haplotypes that differ in *disorder* at equal mean — one allele carrying a fixed pattern, the other bistable across molecules.

6.1 Long reads change what the model is for

The Ising model is in that paper to solve a problem long reads do not have. A ~ 100 bp bisulfite read observes a *fragment* of a haplotype, so $p(x)$ over N CpGs is never observed and must be inferred by marginalising a parametric model over everything each read failed to see. That inference is what costs `CpelAsm` 48 h on 20 CPUs per sample. A 35 kb ONT read observes every CpG in the region on one molecule, so the likelihood is complete-data and the fit is a small concave problem solved by transfer matrix plus L-BFGS with an analytic gradient in milliseconds (`pattern.fit_chain`). Entropy follows in closed form from the Markov structure,

$$H(X) = H(s_1) + \sum_{n=1}^{N-1} H(s_{n+1} \mid s_n), \quad (15)$$

with no enumeration of the 2^N state space. Every recursion is verified against brute-force enumeration to machine precision.

Two further consequences. First, a fully non-parametric alternative becomes viable that is not viable in bisulfite data: with whole patterns observed, one can compare the haplotypes’ *empirical* read distributions directly. The cheapest version is a two-sample Kolmogorov–Smirnov test on read fractions (`pattern.read_fraction_ks`), which — unlike the location tests of Path B — is sensitive to equal-mean disorder imbalance. Second, CPEL can only test where a SNP cluster is; long reads phase whole chromosome arms, so every region is testable.

7 Null distributions

7.1 Yes: whole-read permutation of haplotype labels

The reference null is exactly what the user asked about. `null.split_reads` pools the two haplotypes’ reads and re-splits them at the observed sizes, moving **whole molecules** between haplotypes. Because a read moves intact, within-molecule co-methylation is preserved exactly under H_0 — which is why this is valid where a count-level parametric bootstrap (`P04_permutation_null.py`, which resamples from already-aggregated x/n) is not. `readlevel.test_region_perm` is the per-region version; it is correctly calibrated at every co-methylation level tested (type I 0.042/0.033/0.009 against nominal 0.05).

7.2 The floor that makes a permutation test useless genome-wide

A per-region permutation test with B shuffles has $p \geq 1/(B + 1)$. CPEL’s own bootstrap has the same structure with $p = (1 + \#\{t_\ell \geq t^*\})/(L + 1)$, $L \geq 1000$. Under BH across n regions, a region can only be rejected at rank k if $p \leq \alpha k/n$, so a p floored at $1/(L + 1)$ requires

$$k \geq \frac{n}{\alpha(L + 1)}. \quad (16)$$

For CPEL’s $n = 715,155$ haplotypes at $\alpha = 0.05$ with $L = 1000$, that is $k \geq 14,289$: nothing is callable unless fourteen thousand tests tie at the floor simultaneously, and every discovery then carries an identical p -value and is internally unrankable.

7.3 The fix: stratified pooling plus tail extrapolation

`null.MatchedNull` makes two changes.

Pool across regions within a stratum. Under H_0 a region’s null statistic distribution depends on the region’s *shape*, not its identity, so null draws from comparable regions are exchangeable and can be pooled. This buys $L \sim 10^5$ instead of 10^3 for the same compute. Which variables define “comparable” is the whole argument: CPEL conditions on the number of CpGs alone and generates every null draw at the global minimum coverage $C_{\min} = 5$ regardless of a region’s real depth. Here the strata are (n_cpgs, depth, methylation level). Depth matters for the obvious reason; methylation level matters because every one of these statistics has a null spread that collapses as $\mu \rightarrow 0, 1$ — the same mean-dependence that broke the dispersion estimator in §2.4.

Extrapolate the tail. Fit a generalised Pareto distribution to exceedances over a high threshold u ,

$$\Pr(T > u + y \mid T > u) \approx (1 + \xi y / \sigma)^{-1/\xi}, \quad (17)$$

and read p -values off the fitted tail below the empirical floor. This is the standard peaks-over-threshold treatment of permutation p -values (Knijnenburg et al., *Bioinformatics* 2009), not an invention here.

A failure this introduced, and how it hid. The first implementation trusted the fitted tail without limit and returned $p = 10^{-300}$ from a 60,000-draw null — *including on H_0 data*. Type I error at $\alpha = 0.05$ (0.049) and $\alpha = 0.01$ (0.010) looked correct; only the $\alpha = 10^{-3}$ row (0.0016) and the minimum p gave it away. Every statistic here is bounded in $[0, 1]$, so a fitted GPD with large positive shape — a polynomial tail with no upper endpoint — is structurally misspecified, and thin strata produced exactly that. The fix rejects pathological shapes, requires 200 exceedances, and caps extrapolation at $100\times$ the empirical floor. Post-fix type I error is 0.047/0.0088/0.00075/0.000 at $\alpha = 0.05/0.01/10^{-3}/10^{-4}$, with $p \approx 10^{-7}$ still reachable, which is what BH across 10^5 regions needs.

The general lesson is worth stating separately: the $\alpha = 0.05$ row of a calibration table is not evidence about the regime BH actually operates in. This is the same class of error as the original critique’s finding that type I inflation was $2.9\times$ at $\alpha = 0.05$ but $11\times$ at $\alpha = 0.001$.

8 The generative simulators

The user asked whether the simulations use a generative model with loci labelled `is_asm` and associated parameters, and whether they produce high-fidelity long-read data. The answer to the first is yes; the answer to the second is *qualified, and the qualifications are the interesting part*.

8.1 Generation 1: two-layer mixture, independent CpGs

`simulate_mixture` draws, per locus or region, a Bernoulli ASM label at prevalence π (default 0.005, the deCODE validated rate) and, if true, an effect size $\delta \sim \text{Uniform}(0.2, 0.6)$ applied as $\mu_1 = \mu_0 + \delta/2$, $\mu_2 = \mu_0 - \delta/2$. This is exactly the structure imagined in the question.

Its fatal flaw for this application: it draws each CpG’s depth *and* counts independently inside a per-CpG loop. That is not what long reads do, and it means **no benchmark built on it can detect the design effect**, because it hard-codes $\text{DE} = 1$. Every number in the original results table came from this simulator; all of them are superseded.

8.2 Generation 2: shared molecules

`simulate_reads` draws reads *once per region*, spanning every CpG in it, and adds two parameters the first generation lacked: a co-methylation probability, and a realistic bimodal methylome (70% high / 20% low / 10% intermediate) instead of $\mu = 0.5$ everywhere. It also fixed a quiet bias — clipping $\mu_{1,2}$ to $[0.001, 0.999]$ after adding $\delta/2$ meant the realised effect silently differed from the recorded truth at extreme baselines; the effect is now shrunk to fit instead.

8.3 Generation 3: two ASM axes, three correlation models

Generation 2 is still not adequate for comparing against a pattern-based method, for two measured reasons.

Its correlation is exchangeable, not distance-decaying. Each CpG’s state is copied from a single molecule-level latent with probability q , giving $\text{Corr}(X_j, X_k) = q^2$ for *every* pair regardless of distance. Measured at $q = 0.6$, lags 1–11: 0.361, 0.360, 0.360, 0.359, ... — flat, against a 1-D Ising chain’s 0.600, 0.364, 0.218, 0.131, ... Two consequences: the decay curve of §4 has nothing to measure in it; and under exchangeability the per-read fraction is a *sufficient statistic* for the read, so a pattern method is guaranteed by construction to find nothing that Path B does not.

It has only one ASM axis. Both haplotypes get the same co-methylation, so entropy imbalance has zero true positives by construction.

`simulate_patterns` therefore adds a second axis — haplotypes differing in within-read correlation at *exactly equal mean*, one allele fixed and the other bistable, which is polymorphic/stochastic ASM — with independent prevalence, and three correlation models chosen so that no single one can rig the comparison:

model	correlation structure	favours
epiallele	mixture of discrete methylation patterns	<i>neither</i> (headline)
markov	distance-decaying = 1-D Ising reparameterised	CPEL (correctly specified)
exchangeable	the generation-2 model	Paths A/B

Benchmarking on data drawn from a method’s own model tells you nothing, in either direction; this is why all three are run and why the headline scenario is one neither method assumes.

8.4 Is it high-fidelity? No — and the real data says exactly how not

Three honest limitations, the third only discoverable once real data arrived (§10):

1. The **epiallele** scenario, built to favour neither method, turns out to have an *infinite* fitted decay length: its latent patterns span the whole region, so it is non-parametric but still exchangeable. Only **markov** carries genuine distance decay.
2. Phasing is assumed perfect. Switch errors are simulated (`switch_error_rate`) but modelled in no test.
3. **molecules_markov** assumes a **single exponential decay**, and **real data refutes that** — see §10. The simulator is misspecified against the correlation structure it exists to represent, which is the same class of error that made generation 2 unusable for the CPEL comparison.

9 What is and is not FDR-controlled

9.1 Against Fisher

On generation-1 data with known truth, at BH $\alpha = 0.05$:

depth	method	called	empirical FDR	power
18×	Fisher	365	0.822	0.168
18×	beta-binomial, per-CpG	1	0.000	0.003
18×	beta-binomial, region	45	0.111	0.615
5–40×	Fisher	1256	0.898	0.274
5–40×	beta-binomial, region	56	0.107	0.649

Fisher is not merely underpowered, it is unsafe: 77–90% of its calls are false against a known generative model. Region-level effect-size bias (+0.01–0.04) is also far better than Fisher’s systematic overestimation (+0.19–0.25).

9.2 The edge case: FDR control degrades with the design effect

This is the honest answer to ”the FDR isn’t properly controlled in certain edge cases”. On generation-2 data (40,000 regions, $\pi = 0.01$, bimodal methylome, shared reads), the failure is not sporadic — it is a smooth function of one measurable quantity:

design effect	method	called	FDR	power
1.01	pooled read×CpG	226	0.040	0.565
2.40	pooled read×CpG	129	0.070	0.312
5.49	pooled read×CpG	105	0.695	0.083
1.01	read-level	108	0.000	0.281
5.49	read-level	0	—	0.000
any	whole-read permutation	—	type I 0.042/0.033/0.009	—

Reading this honestly: when CpGs really are independent, pooling is correct *and* more powerful — Path A is not wrong in its own simulation, it is wrong about long-read data *if* the design effect is large. The read-level path is conservative everywhere and never breaks. The permutation null is correctly calibrated throughout and is the reference standard.

So the pre-conditions for quoting an FDR number were: measure the design effect, then choose the path. That measurement had never been done.

10 Real data

10.1 What the input actually looks like

`modkit extract` emits one row per (read, position, modification code), which raises two issues that broke a first implementation and are worth recording because neither is visible from documentation alone.

It is per-strand and does not combine. A CpG occupies $(p, p+1) = (C, G)$ on the plus strand, so its minus-strand C sits at $p+1$ and the *same physical CpG* is reported at p from plus-strand reads and $p+1$ from minus-strand reads. Keying columns on the raw reference position splits every CpG into two columns 1 bp apart sharing zero reads. On HG00146 chr15 the split is 4.89M plus / 5.17M minus, so this is not an edge case. Empirically, normalising minus-strand positions by -1 raises the both-strand agreement from 0.018 to 0.375, confirming the direction; a residual 5.9% of adjacent CpGs at exactly 1 bp — structurally impossible, since a CpG at p makes $p+1$ a G — is repaired by merging.

A 0.5 cutoff on one modification code is not a call. Each CpG appears twice, as **m** (5mC) and **h** (5hmC), with $\text{Pr}(\text{canonical}) = 1 - p_m - p_h$. A real row from this cohort reads $p_h = 0.686$, $p_m = 0.314$: almost certainly modified, yet a $p_m \geq 0.5$ rule scores it a confident *unmethylated*. By §4.2 this injects exactly the error that attenuates the design effect. The extraction now takes the $\arg \max$ over $\{C, m, h\}$, requires it to clear a confidence threshold, and emits a **no-call** otherwise. On HG00146 chr15, 13.2% of positions fall below a 0.80 threshold and 1.0% are 5hmC-dominant; the residual $\varepsilon \approx 0.043$ still attenuates correlations by 0.836.

10.2 The correlation function

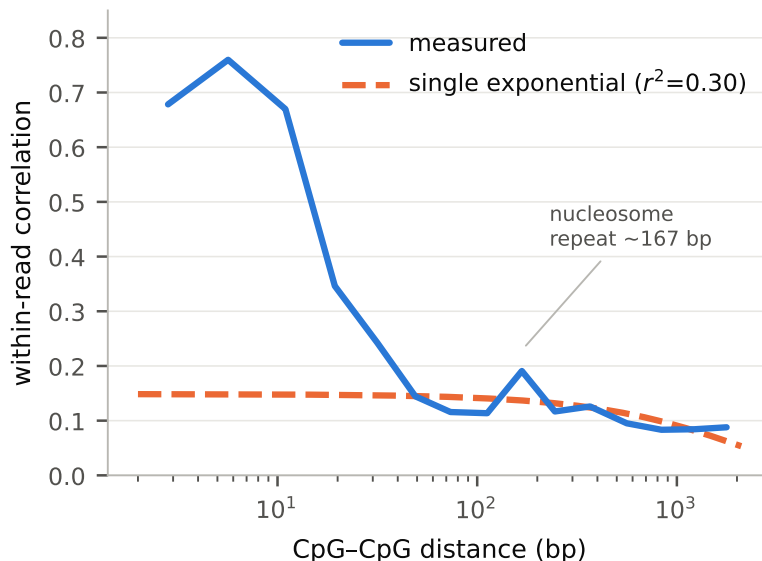


Figure 1: Within-read methylation correlation against CpG-CpG distance, HG00146 chr15 (2,323 windows of 2 kb, ~ 2.5 Mb, call-error corrected). The single-exponential model assumed by `simulate_patterns.molecules_markov` and by `decay.comethylation_decay`’s fit does not describe the data ($r^2 = 0.30$, against > 0.95 on simulation).

Figure 1 is the first measurement of this quantity on the project’s data, and it refutes the model used to simulate it. The structure is **two-component**: correlation peaks at 0.76 around 6 bp, falls steeply to 0.15 by 49 bp — a short-range decay length of order 20–25 bp — and then *plateaus* at 0.07–0.09 out to at least 1775 bp rather than continuing to decay. A local bump at 167 bp coincides with the nucleosome repeat length; on one donor and one bin that is suggestive, not a finding.

10.3 The design effect, measured

window	median DE	IQR	90th pct	% DE > 2	% DE > 5
500 bp (the caller’s scale)	1.29	1.05–1.72	2.20	15%	0.7%
1000 bp (<code>max_span</code>)	1.48	1.17–2.06	2.78	27%	1.5%
2000 bp	1.89	1.43–2.84	3.91	—	4.3%

Against the FDR-versus-DE curve of §9, this says:

Region pooling is broadly defensible at the scale this caller actually uses.
The regime where it collapses is under 1% of regions, not the typical case.

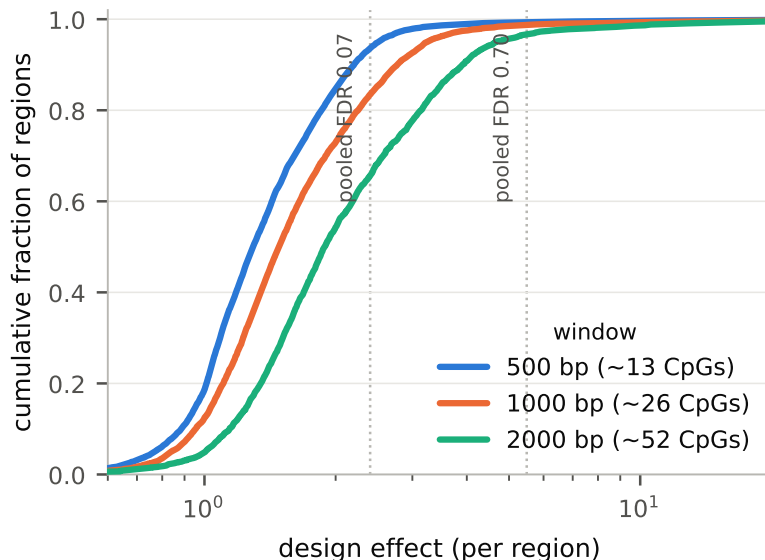


Figure 2: Distribution of the per-region design effect, `readlevel.design_effect` on HG00146 chr15, at three window sizes. Dotted lines mark the design effects at which the pooled test’s FDR was measured in simulation. The caller’s own regions are capped at 1000 bp.

This substantially demotes the read-level and pattern paths. The critique that motivated them measured type I inflation at co-methylation 1.0, i.e. $DE = 5.5$, which is the *99th percentile* of real regions rather than the typical one. Paths B and C remain correct and worth having; the case for making either the default is much weaker than it appeared. The defensible architecture is **design-effect-aware**: pool where the measured DE is low, fall back to the read-level path where it is not. Since DE is now measurable per region, that is a switch, not a research programme.

10.4 An unresolved disagreement

§3 argued that having two independent estimators of the design effect is a deliberate tripwire. It is currently tripped. At 2kb they agree (pooled long-range correlation 0.0733 versus median per-window implied ICC 0.0728); at 500 bp they do not (`implied_design_effect` 2.77 against a measured median of 1.29). The likely cause is a summary mismatch — the correlation curve is pooled with variance weighting, so high-variance windows dominate it, while the design effect is a median over windows, and at short range the across-window spread is much larger. That is a hypothesis, not a demonstration. **Until it is resolved, the direct per-region measurement is the one to use**; it does not depend on the decay-curve machinery at all.

11 Open problems

1. **Resolve the design-effect estimator disagreement** (§10). The tripwire exists to be believed.
2. **Replace the single-exponential decay model** in both the fit and the simulator with a two-component form (short exponential plus floor). The simulator being misspecified against real correlation structure is the recurring failure mode of this project.

3. **Switch errors are modelled in no test.** They move reads between alleles, which attenuates T_{MML} and *inflates* T_{NME} , since a mislabelled allele looks bistable. Two cohort donors carry $\hat{\rho}$ five to ten times everyone else's with essentially zero significant regions; that signature — elevated T_{NME} with normal T_{MML} — would be close to a positive identification, and it is a genome-wide aggregate, so it does not need the per-region power the entropy axis lacks.
4. **The entropy axis may not be reachable.** Maximum power on equal-mean disorder imbalance across every method and scenario tested was 0.034; T_{NME} scored 0.000 on that axis in all three scenarios while retaining power on the mean axis, i.e. at 10–30 reads it largely tracks the mean. The identifiability measurement explains why: at 400 reads the MLE recovers $(\alpha, \beta) = (1.2, 0.6)$ as $(1.26, 0.58)$; at 20 reads it returns $(1.81, 0.10)$.
5. **No haplotype-specific real-data validation yet.** Everything in §10 is unphased, because co-methylation along a molecule is not a haplotype quantity. Every ASM test remains blocked on haplotagged BAMs.

Software

https://github.com/terencewtli/ont_asm_caller. Paths A, B and C are `model.py/region.py`, `readlevel.py`, and `pattern.py`; nulls are `null.py`; correlation measurement is `decay.py`; simulators are `simulate_mixture.py`, `simulate_reads.py` and `simulate_patterns.py`. The figures here are reproduced by `docs/paper/make_figures.py`.